Choosing Image Features for Local Problems: Six Digital Image Processing Cases from Indonesia

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Abstract*—The examples in many image processing literature often use Western environment like faces in their databases and car plates with standard format. This paper reviews six image processing problems which are relevant to Indonesia and which can be addressed using digital image processing techniques, namely disease detection in rice leaves, oil palm fruit ripeness classification, batik motif recognition, Javanese handwritten character recognition, road damage detection, and CCTV under low-light conditions improvement. The six cases will be categorized in three groups: agricultural, heritage, and public infrastructures. For each case, the problem, the image processing technique applied in previous researches, the relevant feature used, and its performance will be discussed. It is discovered that the best feature depends on one single issue: *What kind of visual property changes among different classes?

***Index Terms*—digital image processing, feature extraction, GLCM, color features, zoning, batik, agriculture, low-light enhancement**

I. Introduction

There are many problems in Indonesia that can be solved faster and more cheaply using camera and computer. Indonesia's farmers' harvest can be wasted due to diseases in the plants, palm oil mills require sorting of the fruit bunches daily, batik and Javanese script should be digitally preserved, and roads and other areas require monitoring for any signs of damage and other problems. Most of these tasks are carried out manually and are therefore very time-consuming.

Automating them could be easily achieved with digital image processing (DIP). A typical DIP algorithm consists of the following five stages: image acquisition, image cleaning (preprocessing), segmentation, feature extraction and classification [1]. This paper will examine the fourth step in more detail. The selection of features is usually the key factor in success or failure of the whole process; a good classifier can only distinguish the classes when the features are indeed different enough.

In section II a brief overview of the types of features used in this paper is provided. Six case studies are provided in Section III, organized into three themes. Section IV provides comparison of the cases and Section V is the conclusion.

II. Feature Types Used in This Paper

A. Color Features

Color features are taken from the pixel values of each channel, usually after converting RGB into a space that separates color from brightness, like HSV or L*a*b*. Common choices are the mean, standard deviation, and histogram of each channel. They work when classes differ mainly in color.

B. Texture Features (GLCM)

The Gray-Level Co-occurrence Matrix (GLCM) counts how often a pixel with gray level i appears next to a pixel with gray level j at a given distance and angle [2]. From the normalized matrix $P(i, j)$, several numbers are computed:

$$\text{Contrast} = \sum_{i,j}(i-j)^2 P(i,j) \tag{1}$$

$$\text{Energy} = \sum_{i,j} P(i,j)^2 \tag{2}$$

$$\text{Entropy} = -\sum_{i,j} P(i,j)\log P(i,j) \tag{3}$$

High contrast means sharp changes between neighbors, high energy means a uniform or repeating pattern, and high entropy means a random, complex pattern. Computing GLCM at 0°, 45°, 90°, and 135° captures patterns in different directions.

C. Shape and Layout Features

Shape features describe the outline of a region (area, perimeter, circularity). *Zoning* is a layout feature: a binary character image is divided into an $n \times n$ grid, and the share of black pixels in each cell becomes one feature. A 8×8 grid gives a 64-number vector that describes where the strokes are.

D. Histogram Statistics

The mean, standard deviation, and entropy of the gray-level histogram characterize the brightness, contrast, and texture of the image. These features can be used to measure the quality of images rather than to recognize objects.

III. Cases

Theme 1: Agriculture

A. Rice Leaf Disease Detection

Problem. Rice is Indonesia's main food crop, and diseases such as bacterial leaf blight, brown spot, and leaf smut can

spread quickly. Farmers often cannot tell these diseases apart early.

Processing. Azim *et al.* [3] first removed the background using a threshold on the saturation channel, then isolated the sick area using a threshold on the hue channel. Features were taken only from the sick area, not the whole leaf.

Suitable features. Each disease leaves spots with a different *color* (brown, yellow, dark), a different *shape* (long streaks vs. round spots), and a different surface *texture*. Combining all three gave 86.58% precision with an extreme gradient booster classifier on the UCI rice leaf dataset [3]. The key idea is that cutting out the sick area first makes the features much cleaner.

B. Oil Palm Fruit Ripeness Grading

Problem. Indonesia is the world's biggest palm oil producer. Fresh fruit bunches must be harvested at the right ripeness, because unripe bunches give less oil and overripe bunches raise free fatty acid levels. Graders currently judge ripeness by eye.

Processing. Images of the bunches are captured under controlled lighting, and the fruit region is segmented from the background.

Suitable features. Ripeness is mainly characterized by a change in the color from black or dark purple to reddish orange. Thus, *color features* would be the best choice. Fadilah *et al.* [4] fed the color features to an artificial neural network, and found that the reduction of features using PCA increased the accuracy even more. Texture and shape provide little additional information since all bunches, regardless of the ripeness level, are similarly shaped.

Theme 2: Cultural Heritage

C. Batik Motif Classification

Problem. Indonesia has thousands of batik motifs, such as Parang, Kawung, and Megamendung, each tied to a region and meaning. Classifying them automatically helps digital archiving and protects motifs from wrong labeling.

Processing. Images are converted to grayscale and reduced to fewer gray levels (for example 16) so the GLCM stays small.

Suitable features. Motifs in Batik have repeating *patterns* rather than colors because the same motif can exist in different colors. This gives the correct attribute to be used as *texture*. The authors of [5] calculated energy, entropy, contrast and correlation attributes using GLCM at four different angles, resulting in 16 features. Using cross validation, the Support Vector Machine method produced an accuracy of 92.3% whereas k nearest neighbor produced only 78.3%. Using four different angles is important because different motifs have different dominant directions; for example, Parang motif consists of diagonal lines while Kawung motif consists of circles.

D. Handwritten Javanese Script Recognition

Problem. Aksara Jawa is a traditional script still taught in schools in Central and East Java. Recognizing handwritten characters can help digitize old manuscripts and support learning apps.

Processing. Each character is converted to a binary image, cropped to its bounding box, and resized to a fixed size so all characters can be compared.

Suitable features. Characters are black strokes on white paper, so color and texture carry almost no information. What matters is *where the strokes are*. Handayani *et al.* [6] used *zoning* features with 64 inputs into a backpropagation neural network and reached 77.00% accuracy. The lower accuracy compared to the other cases shows how hard handwriting is: Javanese characters have many similar shapes, and each person writes differently.

Theme 3: Public Infrastructure and Safety

E. Road Damage Detection

Problem. Potholes and cracks are a real danger, especially for the many motorcycle riders on Indonesian roads. Road agencies need a cheap way to find and classify damage over long distances.

Processing. A camera on a vehicle records the road, and each frame is converted to grayscale before feature extraction.

Suitable features. A smooth road has an even, uniform texture, while damage breaks this pattern. Sartika [7] used GLCM features (dissimilarity, correlation, contrast, energy, and angular second moment) with an SVM to sort road images into five damage types. The overall accuracy was 80%, with F1 scores of 0.95 for potholes and 0.89 for cracks. Potholes are easiest to detect because they create a strong, dark break in the texture.

F. Low-Light CCTV Enhancement

Problem. Many CCTV cameras in Indonesian streets, campuses, and boarding houses (*kos*) are low-cost models. At night their images are dark and noisy, so faces and plates cannot be seen. This case is also the topic of the author's ongoing group project on pixel-based enhancement.

Processing. Pixel-based methods such as gamma correction, histogram equalization, and Contrast Limited Adaptive Histogram Equalization (CLAHE) [8] brighten dark areas. Guo *et al.* [9] review these classic methods alongside newer learning-based ones. Classic methods are fast and need no training data, which suits cheap CCTV hardware.

Suitable features. Unlike the other cases, the features here are not used to recognize an object. They are used to *measure* the image and to *tune* the method. The histogram mean decides how strong the gamma correction should be, the standard deviation shows the contrast, and entropy shows how much detail is visible. After enhancement, metrics like PSNR and SSIM compare the result to a reference image.

IV. Comparison and Discussion

Table I summarizes the six cases. Several lessons stand out.

Pick the feature that matches the change between classes. Ripeness changes color, so color wins. Batik motifs change pattern but not color, so texture wins. Characters

TABLE I
SUMMARY OF CASES, SUITABLE FEATURES, AND REPORTED RESULTS

Case	Main Feature	Details	Result
Rice leaf disease [3]	Color + shape + texture	Hue/saturation segmentation, features from sick area	86.58% (XGBoost)
Oil palm ripeness [4]	Color	Color features + PCA	PCA improved accuracy (ANN)
Batik motif [5]	Texture	GLCM, 4 stats $\times$ 4 angles	92.3% (SVM)
Aksara Jawa [6]	Layout	Zoning, 64 features	77.00% (ANN)
Road damage [7]	Texture	GLCM, 5 stats	80% (SVM)
Low-light CCTV [9]	Histogram stats	Mean, std, entropy; PSNR/SSIM	Quality, not class

change stroke position, so zoning wins. Using a feature that does not follow the real difference, like color for batik, only adds noise.

GLCM is the most reused tool. It appears in batik, road damage, and partly in rice disease. It works for any problem where the surface pattern carries the meaning, and it is cheap to compute.

Segmentation before features helps a lot. In the rice case, taking features only from the sick area instead of the whole leaf gave cleaner features. The same idea applies to palm fruit and road images.

Fewer, better features can beat many features. In the palm oil case, PCA reduced the number of features and still raised accuracy [4].

Finally, the reported accuracies (77–92%) show that hand-crafted features are good but not perfect. Deep learning can learn features on its own, but it needs large labeled datasets, which are rare for local topics like Aksara Jawa or regional batik. For these topics, hand-crafted features remain a practical choice.

V. CONCLUSION

In the present paper, six problems arising in Indonesia have been discussed that could be addressed using digital image processing technology. For all of them, the most appropriate feature turned out to be the one that connects to the difference between classes in nature: color – in case of oil palm maturity assessment; texture, when analyzing batik patterns and road surface damages; color, shape, and texture when recognizing diseases on rice plants; layout of strokes, in recognition of Javanese script characters; histogram statistics, in improving images in low light conditions.

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